

JT_DEV_B52_2.1

PHP framework App

1 Initial situation

- 1.1 To build real-world applications, using « vanilla » PHP is rarely efficient, especially for complex projects. We tend to prefer using PHP frameworks. A framework provides a structured foundation and reusable components that speed up development. Moreover, frameworks help us to write code that is more secure and easier to maintain.
- 1.2 There are many PHP frameworks, as today, the most popular are [Symfony](#) and [Laravel](#). In this training project, you will have the opportunity to choose the framework you want to learn : Symfony or Laravel. But remember that the goal is to learn how to use a PHP framework. One day you will have to be able to adapt to other frameworks as well.
- 1.3 Some other existing PHP frameworks are : CodeIgniter, Zend Framework, CakePHP, Yii, Phalcon...
- 1.4 To learn the framework usage, you will create a simple Airbnb clone. The app will allow users to propose their own apartments for rent, and to book apartments proposed by other users.
- 1.5 **Excerpt of app features :**
 - User registration, login and logout with username, email and password (using framework built-in authentication system)
 - Page with all apartments available on the platform
 - Filter list to show only apartments proposed by the connected user (my apartments)
 - Filter list by apartments booked by the user (my bookings)
 - Sort and filter apartments by different criteria (price, location, number of rooms)
 - Users can delete their apartments
 - Page to propose a new apartment for rent
 - Page to edit an apartment (can be done only by the apartment owner or the admin)
 - Page to see the details of an apartment
 - Users can book the apartment (if available)
 - Users can cancel booking
 - Homepage with 3 most rented locations, 2 apartment suggestions for each location
 - Page to edit user profile (username, email, password)
 - Admin dashboard to manage users (list all users, delete users, promote users to admin)
- 1.6 Before diving into code, you will have to design the database schema and propose an app wireframe.

2 Skills from the ICT training plan worked on

Skills ranked by importance of implementation in the project: [consult the training plan](#).

- 2.1 g5 : Implémenter les applications et les interfaces selon le concept en respectant les exigences de sécurité.
- 2.2 c2 : Mettre en œuvre des modèles de données dans un dispositif de stockage de données numériques.
- 2.3 g2 : Vérifier la faisabilité technique des prototypes d'interfaces utilisateur et les développer.
- 2.4 g3 : Évaluer et documenter la sécurité des applications et des interfaces.
- 2.5 g6 : Vérifier la qualité et la sécurité des applications et des interfaces.
- 2.6 a4 : Planifier les projets ICT et les tâches selon un modèle de procédure.
- 2.7 c1 : Identifier et analyser les données et développer des modèles de données
- 2.8 h3 : Implémenter le processus de livraison des applications.



3 Deliverables

- 3.1 GitHub repository containing the working app.
- 3.2 GitHub project with all tasks completed.

4 Andragogic conditions

- 4.1 Work is carried out independently, on an individual basis.
- 4.2 You are allowed to help each other, share ideas and discuss solutions, but each trainee must provide their own implementation with their own code.
- 4.3 Apprentices have access to all the resources they need, particularly on the web.
- 4.4 Don't hesitate to ask AI for explanations, but avoid asking it for answers. The more you ask for answers, the less you learn.
- 4.5 **Maximum allowed time for the project: 14 full days (7.5 hours per day).** You must manage your time independently. On project beginning, inform your trainer of the start date and time.

5 Steps

- 5.1 Create a new GitHub Project from [the PHP rent app template](#) to your account (Use the «...» then «Make a copy» menu). Select your account to host the project and name the project with your GitHub pseudo, for example `php-rent-<your-username>`. **Do not forget to tick the «Draft issues will be copied if selected» checkbox!**
- 5.2 Ensure you have [PHP](#) and [Composer](#) installed on your computer (install recent versions like PHP 8.5+ and Composer 2+).
- 5.3 Ensure you have [Docker](#) installed on your computer to simplify running the database.
- 5.4 Ensure your PHP installation matches the requirements of the framework you have chosen.
- 5.5 Setup your project by following instructions in the «project setup» task.
- 5.6 Implement all the tasks described in the project.

6 Resources

- 6.1 Object-Oriented Programming : <https://grafikart.fr/formations/programmation-objet-php>
- 6.2 Formation Symfony : <https://grafikart.fr/formations/apprendre-symfony-7>
- 6.3 Setup Docker Symfony : <https://symfony.com/doc/current/setup/docker.html>
- 6.4 Symfony documentation : <https://symfony.com/doc/current/index.html>
- 6.5 Formation Laravel : <https://grafikart.fr/formations/laravel>
- 6.6 Documentation Laravel : <https://laravel.com/docs/12.x>

7 Evaluation criteria

- 7.1 All the tasks from the GitHub project have been implemented.
- 7.2 For each task there is a pull request, with a reviewer and a [close keyword](#).
- 7.3 The review comments have been taken into account in the PR.
- 7.4 All the commits must comply with the [conventional commits](#) standard.
- 7.5 The codebase must be formatted with pint (psr12 style).



- 7.6 The app must work out of the box following Readme instructions.
- 7.7 Your codebase respects the security recommendations (protected against sql injections, XSS, and right password hash storage and verification)
- 7.8 All errors are managed on all pages and conveniently displayed to the user.
- 7.9 The README contains a brief description of the app, setup instructions and a list of features.
- 7.10 The style of the app is clean and modern. Use recent CSS features.
- 7.11 All features work as described in the project tasks.
- 7.12 There is no dead code.
- 7.13 The chosen framework best practices are followed.
- 7.14 Migrations are used to create and update the database schema.
- 7.15 The database schema is well designed, with proper relations between tables.
- 7.16 Sensitive information (like database credentials) is not stored in the codebase but in environment variables.
- 7.17 No use of frontend JS frameworks (like React, Vue, Angular...). Only PHP and framework templating engine (Twig or Blade) must be used for rendering HTML.
- 7.18 Proper validation is done on all user inputs.
- 7.19 No use of deprecated functions or features of the chosen framework.
- 7.20 Dependencies are up to date.

